

FMC-DETR: Frequency-Decoupled Multi-Domain Coordination for Aerial-View Object Detection

Ben Liang, *Graduate Student Member, IEEE*, Hongguang Wei, Yuan Liu, *Member, IEEE*, Bingwen Qiu, Yihong Wang, Xiubao Sui, and Qian Chen

Abstract—Remote sensing object detection is a critical technology for real-world applications such as natural resource monitoring, traffic management, and UAV-based rescue. Detecting tiny objects in high-resolution aerial imagery remains challenging due to weak visual cues and insufficient global context modeling in complex scenes. Existing methods often suffer from delayed contextual interaction and limited nonlinear reasoning, which restrict their ability to effectively refine shallow representations and ultimately lead to suboptimal performance. To address these challenges, we propose FMC-DETR, a frequency-decoupled fusion framework for aerial-view object detection. First, we propose the Wavelet Kolmogorov-Arnold Transformer (WeKat) backbone, which employs cascaded wavelet transforms to enhance global low-frequency structure perception in shallow features while preserving fine-grained details, and further leverages Kolmogorov-Arnold networks for adaptive nonlinear modeling of multi-scale dependencies. Second, we introduce the Multi-Domain Feature Coordination (MDFC) module, which refines cross-scale fused representations through partial-channel spatial, spectral, and structural coordination, thereby strengthening small-object-related feature responses in cluttered scenes. Finally, we design the Compact Partial Fusion (CPF) module, which performs compact multi-branch aggregation with progressive partial refinement to improve feature diversity and multi-scale interaction while preserving stable information flow and reducing redundant perturbation. Extensive experiments across multiple remote sensing benchmarks demonstrate that FMC-DETR achieves state-of-the-art performance and significantly outperforming the baseline detector. Code is available at <https://github.com/bloomingvision/FMC-DETR>.

Index Terms—Remote sensing, object detection, frequency-decoupled fusion, multi-scale feature.

I. INTRODUCTION

Object detection, a cornerstone task in computer vision, has demonstrated profound utility in a multitude of industrial applications, including autonomous driving [1], [2], urban surveillance [3]–[6], and remote sensing [7]–[10]. Propelled by the rapid advances in deep learning, particularly Convolutional Neural Networks (CNNs), modern detection frameworks have

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Ben Liang, Hongguang Wei, Yuan Liu, Bingwen Qiu, Yihong Wang, Xiubao Sui, and Qian Chen are with the School of Electronic Engineering and Optoelectronic Technology, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: benliang@njust.edu.cn; weihg@njust.edu.cn; lyuan90_eo@njust.edu.cn; qbw0315@njust.edu.cn; h@njust.edu.cn; sxh@njust.edu.cn; chenqian@njust.edu.cn).

Xiubao Sui is also with the State Key Laboratory of Extreme Environment Optoelectronic Dynamic Measurement Technology and Instrument, Nanjing 210094, China.

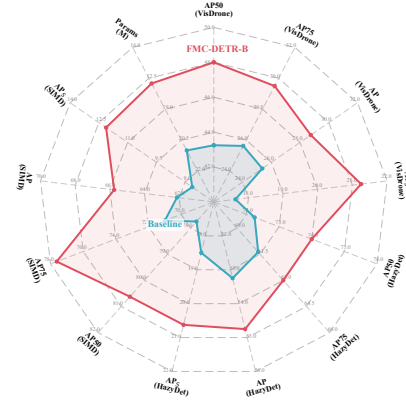


Fig. 1. Performance comparison between FMC-DETR-B and the Baseline [13] on the VisDrone, HazyDet, and SIMD datasets in terms of AP₅₀, AP₇₅, AP, AP_S, and model parameters. The outer regions indicate better performance.

achieved remarkable success on natural images at relatively low resolution. However, these detectors often suffer from significant performance degradation on high-resolution remote sensing imagery, where objects are typically small, weakly featured, randomly distributed, and frequently occluded [11], [12].

To mitigate the challenge of limited appearance information, enhancing the model's perception of global context has emerged as a critical research direction [14], [15]. CNNs, while serving as the default backbone architecture due to their potent local feature extraction capabilities, are constrained by an inherent local inductive bias. Their progressively stacked convolutional layers result in a slowly expanding receptive field, fundamentally limiting their capacity to model the long-range dependencies (LRDs) essential for contextual reasoning. This limitation is particularly disadvantageous for small objects, as distinguishing objects in complex backgrounds requires a comprehensive understanding of the overall scene structure. To transcend the locality of CNNs, researchers have increasingly turned to Transformer-based architectures [16], [17], which leverage self-attention mechanisms to explicitly capture LRDs. However, the computational complexity of Transformers scales quadratically ($O(N^2)$) with respect to the number of input tokens, making their direct application to high-resolution feature maps computationally prohibitive. As a result, a common solution is to adopt a hybrid architecture, where CNNs process high-resolution shallow layers and Transformers are reserved for low-resolution deeper layers. How-

ever, this design introduces a critical performance bottleneck. Shallow feature maps contain rich spatial details and fine-grained structures that are crucial for tiny object detection, but their global structural context is often insufficiently modeled. Delaying global context modeling to deeper layers often causes discriminative small-object cues to be weakened or lost during successive downsampling.

Furthermore, we identify a more fundamental yet often overlooked limitation in how current Transformer models conceptualize contextual integration: they fail to adequately capture the inherently non-linear nature of contextual dependencies. This inadequacy stems from the reliance on Multi-Layer Perceptrons (MLPs) with static, data-agnostic activation functions for feature transformation. In object detection, however, contextual relationships are profoundly non-linear: 1) Strong Scene-Object Priors: The probability of an object's presence can change abruptly with context. For instance, the likelihood of a ship drops to near zero when moving from water to land. Such step-like dependencies cannot be modeled by smooth, linear combinations. 2) Adaptive Cross-Scale Feature Fusion: Fusing semantic information from deep layers with fine-grained details from shallow layers is not a simple weighted average but a dynamic, non-linear process modulated by factors such as object scale and occlusion.

To address these limitations, we propose the FMC-DETR, a frequency-based detector that enhances remote sensing object detection (RSOD) by introducing global context modeling into the high-resolution stages and incorporating adaptive non-linear reasoning into the contextual fusion process. First, the WeKat backbone improves shallow global perception and non-linear contextual modeling through frequency decoupling and asymmetric Kolmogorov-Arnold transformer, while preserving fine-grained high-frequency details. Then, the MDfC module generates an amplitude-guided shallow detail prior from S_2 and refines cross-scale fused features by jointly exploiting spatial responses, edge-aware structural cues, and global spectral interactions, thereby enhancing the discriminability of tiny objects. Finally, the CPF module adopts a structured multi-branch fusion paradigm that combines stable feature preservation with progressive partial refinement, thereby promoting stronger cross-scale feature complementarity and more expressive aggregation. Overall, the proposed framework jointly enhances shallow global awareness, detail-aware cross-scale coordination, and efficient multi-branch feature aggregation, resulting in more robust tiny object detection in complex scenes. Extensive experiments on publicly available remote sensing datasets, including VisDrone [18], HazyDet [19], and SIMD [20], demonstrate the superiority of FMC-DETR over the baseline detector. As shown in Fig. 1, FMC-DETR achieves AP scores of 33.7%, 55.0%, and 66.6% on VisDrone, HazyDet, and SIMD, respectively. The main contributions of this work are summarized as follows:

- We design the WeKat backbone with a heterogeneous split-gating (HSG) mechanism, where HSG-WAVE enhances shallow structure-aware spectral modeling and HSG-AKAT strengthens deep nonlinear semantic abstraction, jointly enabling more expressive multi-stage feature learning.

- We design the MDfC module to enhance small-object detection through multi-domain cross-scale coordination. Specifically, MDfC first converts shallow high-resolution features into an amplitude-guided detail prior, and then jointly refines the fused representation from spatial, spectral, and edge-aware structural perspectives, thereby enhancing tiny-object feature saliency and boundary-aware representation.
- We design the CPF module to address the redundancy and instability of conventional feature aggregation. By combining direct branch interaction, preserved feature pathways, and progressive partial refinement, CPF promotes stronger feature complementarity and more expressive multi-scale representations while maintaining stable information propagation.

II. RELATED WORK

A. CNN-Transformer Object Detectors

CNN and Transformer, as the two cornerstones of the visual domain, have given rise to numerous outstanding detection models. YOLO [21]–[26], a mainstream detection framework primarily built on CNNs, achieves a commendable balance between detection accuracy and inference speed. In parallel, frameworks based on Vision Transformers (ViTs) [17] leverage self-attention mechanisms to continuously advance detection accuracy. In recent years, hybrid architectures combining CNNs and Transformers, such as DETR [27], Deformable DETR [28], RT-DETR [13], D-FINE [29], and DEIM [30], have garnered increasing attention. These models utilize convolutional networks for feature extraction and interaction while employing Transformers for object classification and localization, establishing a new and effective collaborative paradigm. Nevertheless, despite their demonstrated success on public benchmarks like COCO with low-resolution natural images, their efficacy falters when applied to high-resolution aerial imagery. This performance degradation is particularly acute for tiny objects, a shortcoming that can be attributed to their inadequate attention to critical, detail-rich shallow features.

B. Small Object Detection

Detecting small objects has long been challenging. To tackle the persistent challenge of small object detection, recent works have explored various enhancement strategies. GLSAN [14] integrates global-local fusion strategies into a progressive scale-changing network for more accurate bounding box localization. CEASC [31] introduces context-enhanced sparse convolutions to enhance global information, achieving significant improvements in small object detection accuracy. YOLO-DCTI [15] proposes Contextual Transformer (CoT) to integrate global residual and local fusion mechanisms into the detection network, enhancing the contextual utilization of small target pixel information. FBRT-YOLO [11] introduces a Feature Complementary Mapping (FCM) module that integrates semantic and spatial location information, effectively alleviating the loss of small target information and improving small target localization capabilities. Although these methods

TABLE I
CATEGORIZATION OF RELATED STUDIES AND RESEARCH GAPS.

Category	Representative Works	Research Gap
CNN-based detectors	YOLO series [21]–[26]	Limited focus on tiny objects in high-resolution aerial imagery.
Transformer-based detectors	DETR [27], Deformable DETR [28], RT-DETR [13], D-FINE [29], DEIM [30]	Shallow detail features are insufficiently exploited.
Small object detection	GLSAN [14], CEASC [31], YOLO-DCTI [15], FBRT-YOLO [11]	Low-level global context is still underexplored.
Frequency-domain methods	SWANet [32], ELWNet [33], SFS-CNet [34]	Not designed for remote sensing small object detection.
Ours	Proposed method	Jointly models shallow details, low-level global context, and frequency-domain features.

achieved improvements in enhancing small object detection, none of them took into account the importance of capturing global contextual features at lower levels.

C. Frequency-domain Fusion

Frequency-domain features have attracted considerable attention in object detection due to their ability to reveal intricate structural information across diverse scenes. Unlike spatial-domain features typically extracted by conventional convolutions, which capture spatial layout and contextual relationships in an implicit and localized manner, frequency-domain analysis provides a more explicit and global perspective by decomposing intensity variations into distinct spectral components. SWANet [32] uses wavelet transforms to perform multi-scale decomposition of noise and details in images, effectively improving image quality under low light conditions. ELWNet [33] combines the wavelet transform module with CNNs for feature downsampling, using finite parameters to achieve high-quality multi-level encoded features. SFS-CNet [34] proposes a space-frequency selective convolution, which employs a diversion-perception selection strategy to enhance the diversity and uniqueness of features, thereby improving the performance of SAR target detection. Building upon this trend, we explore the integration of frequency- and spatial-domain features specifically for RSOD.

As summarized in Table I, the joint modeling of shallow details, low-level global context, and frequency-domain representations remains underexplored.

III. METHOD

In this section, we introduce the proposed model, named FMC-DETR. As illustrated in Fig. 2, FMC-DETR employs a cascaded hierarchical design, consisting of three main stages from a macro perspective: the backbone network extracts feature maps at various scales, the cross-layer feature interaction structure facilitates the fusion of multi-scale contextual information, and the detection head performs both feature classification and bounding box localization.

A. Wavelet Kolmogorov-Arnold Transformer

In recent years, hybrid CNN-Transformer backbones have shown strong potential for visual representation learning [13], [27], [35], [36]. Yet a stage-wise mismatch remains: shallow high-resolution features often couple under-exploited low-frequency structures with dominant high-frequency details, causing conventional convolutions to favor local textures over stable structural cues. In contrast, deeper semantic stages require selective rather than uniform transformation, since structural information should be preserved while more informative components benefit from stronger nonlinear modeling. To address this issue, we propose the Wavelet Kolmogorov-Arnold Transformer (WeKat), a hybrid backbone built on a Heterogeneous Split-Gating (HSG) design. As shown in Fig. 2, HSG decomposes feature propagation into retention, gating, and computation streams, enabling low-perturbation information preservation, dynamic feature modulation, and controlled strong transformation, respectively. In shallow stages, HSG-WAVE performs wavelet-based frequency decoupling to strengthen low-frequency structural modeling while preserving fine-grained details. In deeper stages, HSG-AKAT equips the computation stream with asymmetric self-attention and Group KAN, enabling more expressive global semantic interaction and selective nonlinear abstraction.

1) *HSG-WAVE*: CNNs exhibit strong locality and translation-equivariant biases, which make shallow layers effective at capturing edges, textures, and other high-frequency patterns. However, these local responses often dominate early representations, while low-frequency structural cues related to object shape, regional consistency, and global layout remain insufficiently exploited. We argue that this weakness stems not only from limited receptive fields, but also from the spectral entanglement between dominant high-frequency details and weaker low-frequency structures. To mitigate this issue, we introduce HSG-WAVE, which explicitly decouples shallow features into low-frequency structural components and high-frequency detail components. By recursively modeling only the low-frequency branch, HSG-WAVE enhances the network’s ability to perceive and exploit global structural cues while preserving fine-grained details.

Given an input feature X_p^c , we first apply the Haar Wavelet Transform (HWT) to decompose it into one low-frequency sub-band and three high-frequency sub-bands:

$$\mathcal{W}(X_p^{c(l)}) = \{X_{LL}^{(l)}, X_{LH}^{(l)}, X_{HL}^{(l)}, X_{HH}^{(l)}\}, \quad (1)$$

where $\mathcal{W}(\cdot)$ denotes the HWT. $X_{LL}^{(l)}$ encodes coarse structural information, while $X_{LH}^{(l)}$, $X_{HL}^{(l)}$, and $X_{HH}^{(l)}$ capture horizontal, vertical, and diagonal details, respectively. For simplicity, we denote the four sub-bands at level l as $\mathcal{S}^{(l)} = \{X_{LL}^{(l)}, X_{LH}^{(l)}, X_{HL}^{(l)}, X_{HH}^{(l)}\}$. To enlarge the effective receptive field (ERF) with low computational cost, HSG-WAVE recursively applies HWT only to the low-frequency branch. Then, at each decomposition level, the four sub-bands are processed by a grouped convolution and then transformed back to the spatial domain to generate a coarser structural prior:

$$R^{(l+1)} = \mathcal{W}^{-1} \left(\mathcal{G}_{5 \times 5} \left(\mathcal{S}^{(l+1)} \right) \right), \quad (2)$$

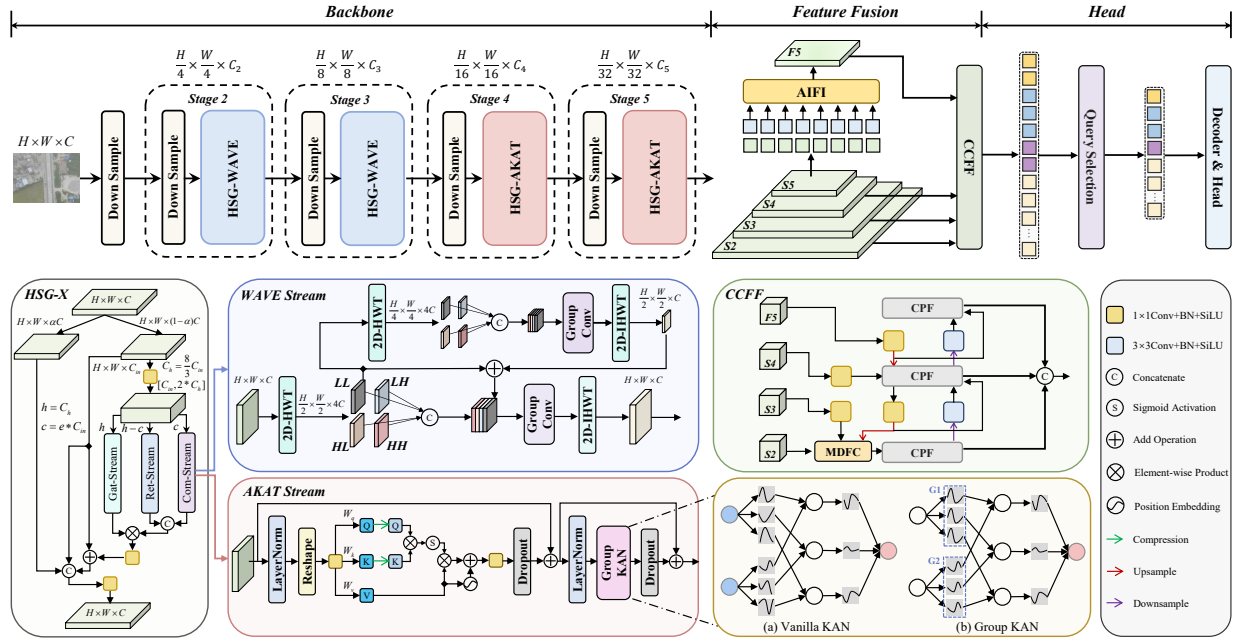


Fig. 2. Overview of the proposed FMC-DETR. The backbone consists of multiple stages to extract and model multi-scale features. Among them, HSG-WAVE and HSG-AKAT are applied to capture correlations in shallow and deep features, respectively. The multi-scale features are then fed into the CCF structure for feature fusion. Finally, the Decoder and Detection Head query the optimized features to produce the final detection results.

where $\mathcal{G}_{5 \times 5}(\cdot)$ denotes the grouped convolution and $\mathcal{W}^{-1}(\cdot)$ denotes the Inverse HWT. Since the recursively decomposed *LL* branch has reduced spatial resolution, structural context modeling introduces only marginal computational overhead. The generated structural prior is used to refine the low-frequency component at the current level, and the enhanced feature is reconstructed as:

$$X_p^{c(l)} = \mathcal{W}^{-1}(X_{LL}^{(l)} + R^{(l+1)}, X_{LH}^{(l)}, X_{HL}^{(l)}, X_{HH}^{(l)}). \quad (3)$$

Through this recursive analysis-and-synthesis process, HSG-WAVE injects global structural semantics into shallow features while retaining fine local details. Compared with conventional convolutions that tend to overemphasize local high-frequency responses, the proposed module improves shallow representations by better coordinating holistic structural perception and fine-grained detail preservation, while suppressing the dominance of unstable high-frequency noise.

2) *HSG-AKAT*: As the network depth increases, the representation focus gradually shifts from structure-detail disentanglement to semantic abstraction and long-range dependency modeling. In the deeper stages of the network, the Transformer's global modeling capacity becomes increasingly valuable. However, standard Transformer blocks suffer from two critical limitations: i) self-attention is computationally expensive and often disrupts the intrinsic 2D spatial structure when flattening features into 1D token sequences; ii) conventional MLPs are static and data-agnostic, which limits their ability to capture complex nonlinear contextual dependencies [37]. Therefore, in the deeper stages, we instantiate the Com-Stream with the Asymmetric Kolmogorov-Arnold Transformer (AKAT), an expressive module designed to enhance global semantic interaction and nonlinear contextual representation.

a) *Asymmetric Self-Attention*: Given an input feature map $X_p^c \in \mathbb{R}^{C \times H \times W}$, we first generate query, key, and value projections by lightweight 1×1 convolutions:

$$Q = W_q X_p^c, \quad K = W_k X_p^c, \quad V = W_v X_p^c, \quad (4)$$

where W_q , W_k , and W_v are learnable projection matrices. To reduce the cost of global interaction while preserving the expressive capacity of the transformed content, we adopt an asymmetric projection strategy, where the dimensions of Q and K are compressed while V remains relatively rich. In this way, the expensive key-query interaction is performed in a lower-dimensional subspace, whereas the value pathway retains sufficient semantic capacity for feature transformation. To maintain spatial awareness, we further derive a dynamic positional bias directly from the value tensor:

$$Pos = \text{DWConv}_{3 \times 3}(V), \quad (5)$$

where DWConv denotes depthwise convolution. The attention output is then computed as

$$\text{Attn}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d}} + Pos\right)V. \quad (6)$$

This design enables efficient global dependency modeling while preserving local spatial inductive cues, making it more suitable for deep semantic refinement within the Computation Stream.

b) *Group KAN*: Instead of employing a conventional MLP after attention, we use a Kolmogorov-Arnold Network (KAN) to enhance nonlinear feature transformation within the deep Com-Stream. KAN replaces static linear mappings with learnable functional expansions:

$$f(x) = \sum_{m=1}^M \beta_m \phi_m(x), \quad (7)$$

where $\{\phi_m(\cdot)\}$ are learnable spline basis functions and β_m are trainable coefficients. To improve scalability, we adopt the Group KAN variant, in which channels are partitioned into G groups and basis parameters are shared within each group:

$$f_g(x) = \sum_{m=1}^M \beta_{g,m} \phi_{g,m}(x), \quad g = 1, \dots, G. \quad (8)$$

Compared with standard MLPs, Group KAN provides a more flexible nonlinear mapping while controlling parameter growth. This is particularly beneficial in deeper stages, where semantic patterns are more abstract and the required feature transformation is less well modeled by a fixed data-agnostic linear projection. By combining asymmetric self-attention with Group KAN, HSG-AKAT enables stronger global semantic interaction and expressive nonlinear transformation within the deep Comp-Stream.

Overall, WeKat establishes a stage-aware representation hierarchy through the HSG design. By separating feature propagation into retention, gating, and computation streams, HSG preserves stable information pathways while selectively enhancing informative channels. HSG-WAVE emphasizes structure-detail coordination in shallow stages, whereas HSG-AKAT strengthens semantic interaction and nonlinear transformation in deeper stages. This progressive design enables WeKat to better balance fine-grained detail preservation, structural consistency, and high-level semantic modeling for high-resolution RSOD.

B. Multi-Domain Feature Coordination

In remote sensing imagery, small targets usually occupy only a few pixels, making shallow high-resolution features crucial for capturing fine-grained local structures [38]. Furthermore, after cross-scale concatenation, the aggregated representation still suffers from substantial heterogeneity, since shallow features mainly provide local details while deeper features contribute stronger semantic context. To this end, we introduce the Multi-Domain Feature Coordination (MDFC) module as a cross-scale coordination block. It couples shallow detail injection with partial-channel multi-domain refinement to calibrate aggregated features from local spatial, global spectral, and edge-structural perspectives, thereby enhancing tiny-object cues while suppressing background interference. As shown in Fig. 3, MDFC is composed of two successive phases: amplitude-guided detail downsampling and multi-domain refinement.

In the first phase, the shallow feature P_2 is transformed into a detail-enhanced prior aligned with the P_3 resolution through an amplitude-modulated fine-grained downsampling unit. Specifically, the input feature is evenly split into a spatial branch X_{p2}^a and a spectral branch X_{p2}^b . The spatial branch is processed by two successive convolutions,

$$F_{\text{spa}} = \text{Conv}_{3 \times 3}^{s=1}(\text{Conv}_{3 \times 3}^{s=2}(X_{p2}^a)), \quad (9)$$

which reduce the spatial resolution while preserving local structural continuity. In parallel, the second branch is fed into an amplitude-modulated spectral pathway. After average

pooling and channel alignment, the feature is then transformed into the frequency domain,

$$F_{\text{fre}} = \mathcal{F}(\phi(\text{AvgPool}(X_{p2}^b))), \quad (10)$$

where $\phi(\cdot)$ denotes a 1×1 convolution. Then, F_{fre} explicitly decomposed into amplitude and phase:

$$A = |F_{\text{fre}}|, \quad P = \angle F_{\text{fre}}. \quad (11)$$

We apply learnable convolutional modulation only to the amplitude,

$$A' = \mathcal{M}(A), \quad (12)$$

where $\mathcal{M}(\cdot)$ denotes the 1×1 convolution operator, while preserving the original phase to maintain structural consistency. The refined spectral feature is reconstructed as

$$F_{\text{am}} = \mathcal{F}^{-1}(A' \odot e^{jP}). \quad (13)$$

Finally, the spatially downsampled response and the amplitude-modulated response are fused by element-wise interaction:

$$F_{p2 \rightarrow p3} = \phi(F_{\text{spa}} \odot F_{\text{am}}), \quad (14)$$

In this way, shallow details are not injected in raw form, but selectively reweighted by frequency-aware amplitude cues before being aligned to the P_3 scale.

In the second phase, the generated shallow detail prior is aggregated with the adjacent multi-scale features:

$$F_{\text{cat}} = \text{Concat}[\text{Up}(Y_4), \phi(P_3), F_{p2 \rightarrow p3}], \quad (15)$$

where $\text{Up}(\cdot)$ denotes the upsampling operator. The concatenated representation is then refined by a multi-domain refinement unit. Within the refinement branch, spatial, spectral, and edge-aware structural responses are jointly modeled and aggregated. First, the feature is split into a process branch F_p and an identity branch F_i , where only a subset of channels is sent to the heavy transformation path, while the remaining channels preserve low-perturbation information flow. A spatial branch captures local patterns by depthwise convolution:

$$F_{\text{spa}} = \text{DWConv}_{3 \times 3}(F_p). \quad (16)$$

A spectral mixing branch performs global frequency-domain interaction. Specifically, we transform F_p into the frequency domain and concatenate its real and imaginary parts:

$$\hat{F}_p = \mathcal{F}(F_p), \quad \hat{F}_{\text{ri}} = [\Re(\hat{F}_p), \Im(\hat{F}_p)]. \quad (17)$$

The concatenated spectral response is then modulated by a learnable 1×1 convolution and mapped back to the spatial domain:

$$F_{\text{freq}} = \left| \mathcal{F}^{-1}(\Gamma(\hat{F}_{\text{ri}})) \right|, \quad (18)$$

where $\Gamma(\cdot)$ denotes spectral channel mixing followed by nonlinear activation. To further inject explicit structural priors, we derive an edge-aware response by using the frequency-enhanced feature to gate Sobel gradients:

$$M_{\text{edge}} = \sigma(\delta(\text{GAP}(F_{\text{freq}}))) \odot \text{Sobel}(F_p), \quad (19)$$

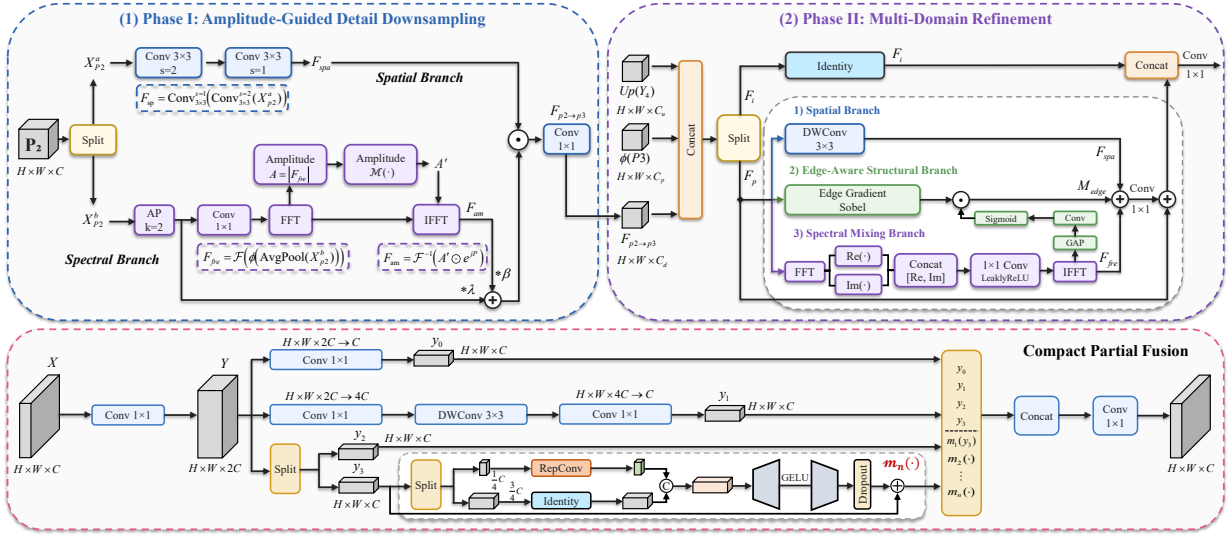


Fig. 3. Overview of the proposed MDFC and CPF modules. The upper part shows the MDFC module, which consists of two successive phases: (1) amplitude-guided detail downsampling and (2) multi-domain refinement. The lower part illustrates the Compact Partial Fusion (CPF) module.

where $GAP(\cdot)$ denotes global average pooling, $\delta(\cdot)$ is a learnable convolution transform, and $\sigma(\cdot)$ denotes the sigmoid function. The three responses are then jointly aggregated:

$$F_{join} = F_p + \phi(F_{spa} + F_{freq} + M_{edge}). \quad (20)$$

Finally, the refined process branch is concatenated with the identity branch and fused by a 1×1 convolution:

$$F_{MDFC} = \phi(\text{Concat}[F_{join}, F_i]). \quad (21)$$

Overall, the MDFC is a cross-scale multi-domain coordination module that first converts shallow low-level features into an amplitude-guided detail prior and then refines the aggregated representation through partial-channel spatial, spectral, and structural modeling. This design is particularly beneficial for small object detection, as it strengthens the fine-grained representation with complementary local detail, global frequency context, and explicit edge cues, while preserving computational efficiency and avoiding excessive disturbance to all channels.

C. Compact Partial Fusion

Effective feature fusion and interaction are essential for improving representation quality, especially in complex scenarios that require both sufficient local detail modeling and stable feature propagation [11], [39], [40]. To enhance feature diversity while avoiding excessive feature perturbation, we propose Compact Partial Fusion (CPF), a compact multi-branch aggregation module with progressive partial refinement.

As illustrated in Fig. 3, given an input feature map $X \in \mathbb{R}^{H \times W \times C}$, CPF first expands the channel dimension through a 1×1 convolution. Based on Y , four complementary feature branches are constructed. The first branch applies a lightweight 1×1 convolution to produce y_0 , providing direct channel interaction. The second branch adopts a pointwise-depthwise-pointwise transformation to generate y_1 , which enhances local

contextual responses with a larger spatial mixing capacity. Meanwhile, Y is split into two parts, denoted as y_2 and y_3 , to preserve the projected features and maintain stable information flow:

$$y_0 = \phi_0(Y), \quad y_1 = \phi_1(Y), \quad (y_2, y_3) = \text{Split}(Y), \quad (22)$$

where $\phi_0(\cdot)$ is a 1×1 convolution and $\phi_1(\cdot)$ denotes the pointwise-depthwise-pointwise branch. To further refine informative responses, CPF progressively processes y_3 with a sequence of partial re-parameterized refinement blocks:

$$M_i = \mathcal{P}_i(M_{i-1}), \quad i = 1, \dots, n, \quad (23)$$

where $M_0 = Y_3$. In each block, only a subset of channels is transformed by a re-parameterized 3×3 convolution, while the remaining channels are preserved through an identity path. The two parts are then concatenated and passed through a lightweight channel transformation with residual connection. This partial refinement strategy improves spatial representation while avoiding unnecessary disturbance to all feature channels. Finally, CPF aggregates the directly transformed branches, preserved features, and progressively refined features through concatenation followed by a 1×1 fusion convolution:

$$\hat{X} = \phi(\text{Concat}(y_0, y_1, y_2, y_3, m_1, \dots, m_n)). \quad (24)$$

Compared with single-path aggregation, CPF introduces a more compact yet diverse feature interaction pattern. The direct transformation branches enhance channel and local contextual modeling, the preserved branches maintain stable feature propagation, and the partial refinement path progressively strengthens informative spatial responses. As a result, CPF improves feature aggregation capability while retaining a stable information pathway, which is beneficial for robust object detection in complex scenes.

IV. EXPERIMENTS

In this section, we first introduce the RSOD datasets used in our experiments, including VisDrone [18], HazyDet

[19], and SIMD [20]. We then describe the experimental settings and present comparisons with state-of-the-art methods. Subsequently, ablation studies are conducted to evaluate the effectiveness of each component, followed by qualitative visualizations for further analysis.

A. Datasets

1) *VisDrone*: VisDrone is a large-scale benchmark specifically designed for object detection in aerial imagery. It consists of high-resolution UAV-captured images from diverse urban and suburban scenes across 14 cities in China, covering a wide range of real-world scenarios. The dataset contains 6,471 training, 548 validation, and 3,190 test images, with over 2.5 million bounding boxes annotated across ten common object categories. Due to the prevalence of small objects, severe occlusions, crowded scenes, and complex backgrounds, VisDrone presents significant challenges for detection algorithms, making it a valuable benchmark for evaluating the robustness and generalization of aerial object detection models.

2) *HazyDet*: HazyDet is the first large-scale benchmark dedicated to object detection in hazy aerial imagery. It contains 11,000 high-resolution images collected from both real-world scenes and physics-based simulations, with 383,000 bounding boxes annotated across three vehicle categories: car, truck, and bus. The dataset is split into 8,000 training, 1,000 validation, and 2,000 test images. By incorporating naturally captured hazy images and synthetic data with varying visibility levels, HazyDet fills a critical gap left by clear-weather benchmarks and provides a valuable resource for evaluating the robustness of detection models under adverse weather conditions.

3) *SIMD*: SIMD is a medium-scale dataset specifically designed for multi-scale and multi-class vehicle detection in satellite imagery. It consists of 5,000 RGB images with a resolution of 1024×768 , collected from 79 locations across Europe and the United States via Google Earth, and adopts a 4:1 training-to-testing split. In total, it provides 45,096 annotated objects spanning 15 categories, mainly vehicles such as cars, trucks, buses, and long vehicles, as well as multiple aircraft types and boats. The diversity in object appearances, scales, densities, and backgrounds makes SIMD a valuable benchmark for advancing aerial object detection, surveillance, and automatic scene analysis under realistic conditions.

B. Implementation Details

All experiments are conducted using PyTorch 2.3 with Python 3.10 on an NVIDIA GeForce RTX 4090D GPU (24 GB). To ensure fair and reproducible comparisons, our method is built upon the RT-DETR [13] baseline and trained from scratch without using any pretrained weights or multi-scale training strategies. Mosaic augmentation is employed during training to improve data diversity and robustness. Unless otherwise specified, the input resolution is fixed at 640×640 during both training and testing. The network is trained for 200 epochs using the AdamW optimizer, with a momentum of 0.9, a weight decay of 0.0005, a batch size of 4, and an initial learning rate of 0.0001. We evaluate performance primarily using the COCO-style Average Precision (AP), along with

additional metrics at different IoU thresholds and object scales, including AP_{50} , AP_{75} , and AP_S .

C. Comparison with state-of-the-arts

1) *Results on VisDrone Dataset*: The results in Table II demonstrate the superior performance of the proposed FMC-DETR on the VisDrone validation set. In particular, FMC-DETR-T achieves the best overall performance, reaching an AP of 33.7% and an AP_{50} of 53.6%, which establishes a new state-of-the-art among all compared detectors. It also maintains a relatively compact model size of 13.8M parameters, showing a favorable trade-off between detection performance and model complexity. Compared with recent YOLO-based detectors, including YOLOv12-L, YOLOv13-L, YOLO26-L, and YOLO-Master-L, FMC-DETR-T consistently achieves higher AP_{50} scores, with gains of 11.6, 13.1, 10.6, and 10.8 points, respectively. Notably, FBRT-YOLO is a recent representative detector specifically developed for aerial small object detection. Compared with its strongest variant, FBRT-YOLO-X, FMC-DETR-T improves AP and AP_{50} by 3.6 and 5.2 points, respectively, while using fewer parameters and lower computational cost. Compared with the Transformer-based baseline RT-DETR-R18, FMC-DETR-T improves AP from 26.7% to 33.7% and AP_{50} from 44.6% to 53.6%, corresponding to gains of 7.0 and 9.0 percentage points, respectively. In addition, the general variant FMC-DETR-B achieves 29.4% AP and 48.4% AP_{50} with 17.4M parameters and 62.9G FLOPs, demonstrating that the proposed design also provides a competitive efficiency-accuracy trade-off.

2) *Results on HazyDet Dataset*: We compare FMC-DETR-B against several representative object detectors on the challenging HazyDet dataset. As shown in Table III, FMC-DETR-B achieves the highest overall AP of 55.0%, surpassing previous strong baselines such as DeCoDet [19] (52.0%), YOLOv12-L (52.6%), and DEIM-D-FINE-S (53.5%). These results underscore the superior detection capability of our method in complex aerial-view scenarios. In terms of category-wise performance, FMC-DETR-B achieves a leading 63.7% AP on the Car class, which dominates the dataset and primarily consists of small, densely packed targets—highlighting the model's effectiveness in capturing fine-grained details under low-visibility conditions.

3) *Results on SIMD Dataset*: We evaluate FMC-DETR-B on the SIMD dataset and compare its performance with several recent object detection models. As shown in Table IV, FMC-DETR-B achieves an AP of 66.6%, outperforming models like RT-DETR-R18 (63.7%), YOLOv8-L (63.1%), and YOLOv9-M (62.2%), while maintaining a relatively low parameter count of 17.4M and FLOPs of 62.9G. Compared with improved models similar to DETR, FMC-DETR-B achieved the highest AP_{50} , leading Deform DETR [28], EMSD-DETR [44], and HPS-DETR [45] by 6.9%, 2.3%, and 3.1%, respectively.

D. Ablation Studies

To validate the effectiveness of our proposed modules, we conducted comprehensive ablation studies across multiple datasets, including VisDrone [18], HazyDet [19], and SIMD [20], as shown in the following tables.

TABLE II

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART DETECTORS ON THE VisDRONE VALIDATION SET. **FMC-DETR-BASE (FMC-DETR-B)** REFERS TO THE GENERAL MODEL, AND **FMC-DETR-T (FMC-DETR-TINYAWARE)** REFERS TO THE ENHANCED VARIANT TAILORED FOR SMALL OBJECTS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN **RED** AND **BLUE**, RESPECTIVELY.

Method	Publication	Input size	AP	AP ₅₀	Param	FLOPs
CNN-Based						
YOLOv8-L [21]	Ultralytics	640×640	28.4	45.9	43.7M	165.2G
YOLOv8-X [21]	Ultralytics	640×640	28.9	46.8	68.2M	257.8G
YOLOv9-M [22]	ECCV2024	640×640	25.1	41.9	20.0M	76.3G
YOLOv10-L [23]	NeurIPS2024	640×640	27.6	44.6	24.4M	120.3G
YOLOv10-X [23]	NeurIPS2024	640×640	28.7	46.1	29.5M	160.4G
YOLOv11-M [24]	Ultralytics	640×640	25.0	42.0	20.1M	68.0G
YOLOv11-L [24]	Ultralytics	640×640	25.5	42.2	25.3M	86.9G
YOLOv11-X [24]	Ultralytics	640×640	26.6	43.8	56.9M	194.9G
YOLOv12-L [25]	NeurIPS2025	640×640	25.1	42.0	26.4M	88.9G
YOLOv13-L [26]	arXiv2025	640×640	24.2	40.5	27.6M	88.4G
YOLO26-L [24]	arXiv2025	640×640	25.7	43.0	24.8M	86.4G
YOLO-Mater-L [41]	CVPR2026	640×640	25.9	42.8	58.4M	138.1G
DTSSNet [6]	TGRS2024	640×640	24.2	39.9	10.1M	49.6G
FBRT-YOLO-M [11]	AAAI2025	640×640	28.4	45.9	7.2M	58.7G
FBRT-YOLO-L [11]	AAAI2025	640×640	29.7	47.7	14.6M	119.2G
FBRT-YOLO-X [11]	AAAI2025	640×640	30.1	48.4	22.8M	185.8G
Transformer-Based						
Deformable DETR [28]	ICLR2020	1300×800	27.1	42.2	40.0M	173.0G
Sparse DETR [42]	ICLR2022	1300×800	27.3	42.5	40.9M	121.0G
RT-DETR-R18 [13]	CVPR2024	640×640	26.7	44.6	20.0M	60.0G
RT-DETR-R50 [13]	CVPR2024	640×640	28.4	47.0	42.0M	136.0G
Mamba-YOLO-T [43]	AAAI2025	640×640	21.0	36.8	6.0M	13.6G
Mamba-YOLO-B [43]	AAAI2025	640×640	23.9	40.8	21.8M	49.6G
DEIM-D-FINE-N [30]	CVPR2025	640×640	17.8	31.5	3.7M	7.1G
DEIM-D-FINE-S [30]	CVPR2025	640×640	24.3	40.6	10.1M	24.9G
FMC-DETR-B (Ours)	-	640×640	29.4	48.4	17.4M	62.9G
FMC-DETR-T (Ours)	-	640×640	33.7	53.6	13.8M	146.8G

TABLE III

COMPARISON OF THE PERFORMANCE OF DIFFERENT SOTA DETECTORS ON THE HAZYDET DATASET.

Model	AP	AP ^{car}	AP ^{truck}	AP ^{bus}	Param
RT-DETR-R18	53.8	63.2	36.0	62.2	60.0M
IAYOLO	38.3	44.1	22.2	48.6	61.8M
MS-DAYOLO	48.3	59.4	28.5	57.0	40.0M
TOOD	51.4	58.4	33.6	62.2	32.0M
Cascade RCNN	51.6	59.0	34.2	61.7	69.1M
Defrorm DETR	51.5	58.4	33.9	62.3	40.0M
DeCoDet	52.0	60.5	34.0	61.9	34.6M
YOLOv12-L	52.6	57.5	37.4	62.6	26.4M
YOLOv13-L	52.7	57.5	37.6	63.1	27.6M
DEIM-D-FINE-S	53.5	62.5	34.0	64.0	10.1M
FBRT-YOLO-X	55.0	59.4	41.0	64.8	22.8M
FMC-DETR-B	55.0	63.7	37.6	63.5	17.4M

TABLE IV

COMPARISON OF THE PERFORMANCE OF DIFFERENT SOTA DETECTORS ON THE SIMD DATASET.

Model	AP	AP ₅₀	Param	FLOPs
RT-DETR-R18	63.7	78.6	20.0M	60.0G
YOLOv8-L	63.1	78.1	43.7M	165.2G
YOLOv9-M	62.2	76.6	20.0M	76.3G
DEIM-D-FINE-S	66.2	82.0	10.2M	24.9G
FBRT-YOLO-X	66.2	83.0	22.8M	185.8G
Deform DETR	59.7	75.6	40.0M	196.0G
EMSD-DETR	64.3	79.4	18.4M	68.3G
HPS-DETR	63.5	79.8	15.5M	68.3G
FMC-DETR-B	66.6	80.9	17.4M	62.9G

1) *Effectiveness of the Proposed Modules Across Datasets:* Tables V and VI present the ablation studies of WeKat, MDFC, and CPF. On VisDrone, all three modules consistently improve the RT-DETR-R18 baseline. Notably, WeKat increases AP from 26.7% to 27.8% while reducing the parameters and FLOPs from 20.0M/60.0G to 16.1M/53.3G, indicating that the proposed backbone improves feature representation with lower model complexity. Among the individual modules, MDFC achieves the largest gain, reaching 28.2% AP and 46.7% AP₅₀, which verifies the importance of multi-domain feature coordination for aerial small objects. CPF also brings consistent improvements, showing that compact partial aggregation is beneficial for enhancing local feature interaction and stable information fusion. By integrating all three modules, the full model achieves the best performance, with 29.4% AP, 48.4% AP₅₀, and 21.4% AP_S, corresponding to improvements of 2.7, 3.8, and 2.9 points over the baseline, respectively. The cross-dataset results on HazyDet and SIMD further validate the generalization ability of the proposed components. On HazyDet, the complete model improves AP/AP₅₀/AP₇₅ from 73.6/53.3/62.2 to 74.5/55.0/63.6 (↑0.9/↑1.7/↑1.4). On SIMD, the gains are more pronounced, with AP/AP₅₀/AP₇₅ increasing from 63.7/78.6/72.4 to 66.6/80.9/77.8 (↑2.9/↑2.3/↑5.4). These consistent improvements across different datasets demonstrate that WeKat, MDFC, and CPF provide complementary benefits and jointly enhance feature representation and detection robustness in aerial object detection scenarios.

2) *Analysis of WeKat Backbone Design:* To further validate the design motivation of WeKat, we conduct three additional

TABLE V
ABLATION STUDIES ON THE VISDRONE DATASET.

WeKat	MDFC	CPF	AP	AP ₅₀	AP _S	Param	FLOPs
×	×	×	26.7	44.6	18.5	20.0M	60.0G
✓	×	×	27.8	46.1	19.5	16.1M	53.3G
×	✓	×	28.2	46.7	19.8	20.2M	61.4G
×	×	✓	27.6	45.4	19.3	20.7M	61.2G
✓	✓	×	28.9	47.6	20.3	16.6M	58.7G
✓	✓	✓	29.4	48.4	21.4	17.4M	62.9G

TABLE VI
ABLATION STUDIES ON THE HAZYDET AND SIMD DATASETS.

WeKat	MDFC	CPF	HazyDet			SIMD		
			AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
×	×	×	73.6	53.3	62.2	63.7	78.6	72.4
✓	×	×	73.8	54.0	62.7	63.9	78.5	73.8
×	✓	×	73.7	54.7	63.1	64.6	79.2	74.3
×	×	✓	73.6	54.3	62.9	65.2	80.1	74.2
✓	✓	×	74.2	54.8	63.5	64.4	78.7	75.3
✓	✓	✓	74.5	55.0	63.6	66.6	80.9	77.8

ablation studies.

First, we examine the necessity of the wavelet-based shallow modeling in HSG-WAVE. Specifically, we keep the overall HSG routing structure and the remaining backbone unchanged, but replace the wavelet-based spectral decomposition in HSG-WAVE with the frequency-domain design adopted in MDFC. As shown in Table VII, the original HSG-WAVE achieves better AP/AP₅₀/AP₇₅ with fewer parameters and lower FLOPs. This is because wavelet decomposition preserves spatial locality while explicitly separating low-frequency structures from high-frequency details, enabling recursive low-frequency refinement to capture broader structural context without interfering with local discriminative responses. By contrast, the FFT-based operation in MDFC emphasizes global spectral interaction, which is more appropriate for cross-scale feature refinement than for shallow structure-aware modeling.

Second, we analyze the sensitivity of HSG with respect to the initial split ratio α and the Com-Stream allocation ratio e . Here, α controls the first channel allocation in HSG, while e determines the relative capacity assigned to the internal Com-Stream. As shown in Table VIII, increasing α from 0.25 to 0.50 brings a clear improvement, indicating that an appropriate channel allocation is important for balancing feature preservation and transformation. Although $\alpha = 0.75$ achieves slightly higher accuracy, it also increases the computational cost from 53.3G to 69.9G. Therefore, we adopt $\alpha = 0.50$ as a better accuracy-complexity trade-off. For the Com-Stream allocation, $e = 1.00$ provides stable overall performance with moderate FLOPs, while further increasing it to 1.25 brings limited gains.

Finally, we further examine the optimization behavior of WeKat by comparing the validation IoU loss and classification loss during training. Fig. 4 shows that, relative to the baseline, WeKat consistently attains lower validation IoU and classification losses throughout training, with a more favorable

TABLE VII
ABLATION ON THE SPECTRAL MODELING BLOCK IN WEKAT. WE REPLACE THE PROPOSED WAVE BLOCK WITH MDFC WHILE KEEPING THE REMAINING BACKBONE DESIGN UNCHANGED.

Spectral block	AP	AP ₅₀	AP ₇₅	Param	FLOPs
WAVE (Ours)	27.8	46.1	28.2	16.1M	53.3G
MDFC	27.7	45.9	27.9	16.3M	55.3G

TABLE VIII
SENSITIVITY ANALYSIS OF THE HSG DESIGN IN WEKAT. α DENOTES THE INITIAL SPLIT RATIO, AND e DENOTES THE COM-STREAM ALLOCATION RATIO. † INDICATES THE DEFAULT SETTING.

α	e	AP	AP ₅₀	AP ₇₅	AP _S	FLOPs
<i>Varying α with fixed $e = 1.0$</i>						
0.25	1.00	26.7	44.6	26.9	19.0	42.0G
0.50†	1.00†	27.8	46.1	28.2	19.5	53.3G
0.75	1.00	28.2	46.8	28.5	20.1	69.9G
<i>Varying e with fixed $\alpha = 0.50$</i>						
0.50	0.75	27.6	46.2	28.0	19.6	51.9G
0.50†	1.00†	27.8	46.1	28.2	19.5	53.3G
0.50	1.25	27.8	46.4	27.8	19.7	55.2G

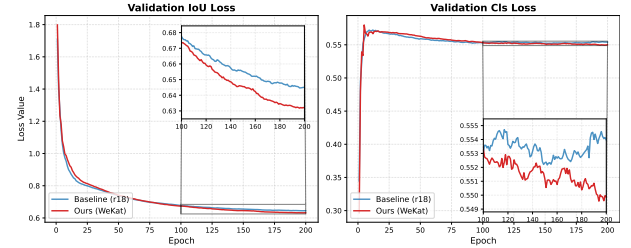


Fig. 4. Comparison of validation loss curves between the baseline RT-DETR and the proposed WeKat.

convergence trajectory in the later epochs. In particular, the gap becomes clearer after the initial optimization phase, where WeKat reaches lower final losses for both localization and classification. This suggests that the proposed HSG routing improves feature transformation and information flow, thereby making the backbone easier to optimize.

3) *Effectiveness of Detection Layer*: In the backbone design, we investigate the necessity of the $S5$ feature layer for object detection. As the number of down-sampling operations increases, the high-frequency details of small objects are irreversibly lost, making the low-resolution $S5$ feature largely redundant for precise localization. As shown in the heatmap visualizations of different detection layers in Fig. 5, the high-resolution $D2$ layer contributes significantly more to small object detection compared to deeper layers. Motivated by this observation, we reduce one downsample stage in the WeKat backbone (equivalent to removing the $S5$ feature layer) and adjust the detection layers to $[D2, D3, D4]$. This modification improves the overall AP from 27.8% to 30.9%, with AP_S increasing by 4.0%, thereby validating the effectiveness of high-resolution features for small object detection. Furthermore, from the perspective of feature redundancy, different

TABLE IX

ABLATION STUDY ON THE EFFECT OF $S5$ BACKBONE LAYER AND DETECTION HEAD DESIGN. “✓” AND “✗” INDICATE THE PRESENCE OR ABSENCE OF THE $S5$ STAGE, RESPECTIVELY.

Model	S5	Detect Layer	AP	AP ₅₀	AP ₇₅	AP _S	Param	FLOPs
Baseline	✓	[D3, D4, D5]	26.7	44.6	26.9	18.5	20.0	60.0
WeKat	✓	[D3, D4, D5]	27.8	46.1	28.2	19.5	16.3	54.7
WeKat	✗	[D2, D3, D4]	30.9	50.2	31.8	23.5	12.9	120.1
WeKat	✗	[D3, D4]	30.8	50.2	31.5	23.3	12.8	103.0
WeKat	✗	[D2, D4]	31.6	50.9	32.9	23.9	12.8	115.8
FMC-DETR-B	✗	[D3, D4, D5]	29.4	48.4	30.2	21.4	17.4	62.9
FMC-DETR-T	✗	[D2, D4]	33.7	53.6	35.4	25.8	13.8	146.8

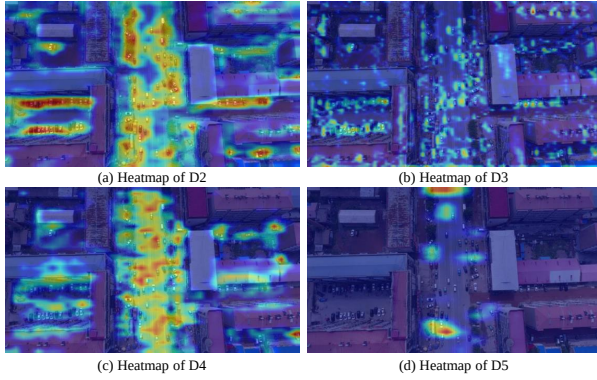


Fig. 5. Visualization of feature activations across detection layers for small objects. Here, [D2 – D5] denote detection layers at different feature scales, where the spatial resolution decreases progressively by a factor of 2 at each stage.

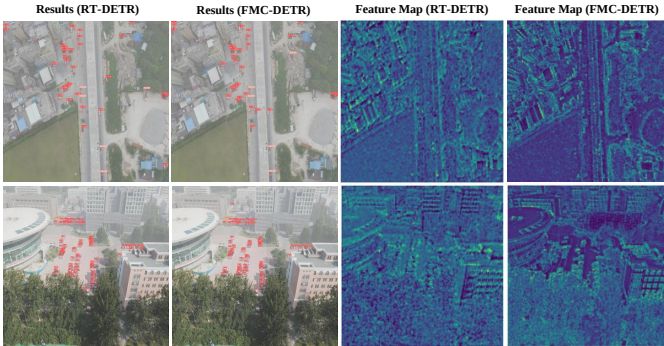


Fig. 6. Visualizations of the detection results of baseline and our proposed method on HazyDet.

detection layers contribute unevenly to objects of varying scales. By combining this analysis with the heatmap results in Fig. 5, we find that the $D2$ and $D4$ layers provide more sufficient and concentrated responses for small objects. Based on this insight, FMC-DETR-T ultimately adopts [D2, D4] as its detection layers, achieving the best performance with 53.6% AP₅₀ and 25.8% AP_S.

E. Visualization Analysis

To further demonstrate the effectiveness of FMC-DETR in aerial-view scenarios, we visualize both feature responses and detection results in Fig. 6 and Fig. 7. Compared with the baseline, FMC-DETR produces clearer structural representations

TABLE X

INFERENCE EFFICIENCY UNDER DIFFERENT BATCH SETTINGS. LATENCY AND FPS ARE MEASURED USING PYTORCH FP32 ON AN NVIDIA RTX 4090 GPU WITH AN INPUT SIZE OF 640×640 .

Method	batch size=1		batch size=16	
	Latency (ms/img)	FPS (img/s)	Latency (ms/img)	FPS (img/s)
RT-DETR-R18	20.8	48.1	4.9	205.0
FMC-DETR-B	56.2	17.8	9.3	107.6

and more informative contextual responses, enabling more accurate object localization. It better captures both holistic object shapes and fine-grained boundary details, resulting in more precise predictions. Furthermore, the qualitative comparisons show that FMC-DETR achieves stronger multi-scale detection performance, particularly for small objects. Relative to RT-DETR, our method localizes small targets more accurately and pays greater attention to their surrounding contextual regions, indicating a more effective use of contextual cues. These observations validate the superiority of FMC-DETR for multi-object detection in complex aerial scenes.

F. Limitations and Discussion

Despite its favorable detection accuracy, the proposed method still has several limitations. First, although our design reduces the number of parameters, it does not lead to a proportional reduction in computational cost. The multi-branch feature coordination and compact feature fusion introduce additional operations, resulting in relatively higher FLOPs and potentially lower inference speed in practice. In addition, as reported in Table X, FMC-DETR-B achieves 56.2 ms latency and 17.8 FPS under bs=1, and improves to 9.3 ms per image and 107.6 FPS under bs=16. FMC-DETR-B is slower than the original RT-DETR-R18 baseline because it intentionally introduces stronger feature representation modules to enhance RSOD in challenging scenes. In particular, the KAN-based component involves more complex nonlinear function modeling than standard convolutional operations, which increases both computational cost and practical inference overhead. Moreover, compared with highly optimized operators such as convolution and matrix multiplication, current KAN-style operations are not yet fully optimized at the low-level kernel and inference-engine level, leading to less favorable runtime efficiency on GPUs. Therefore, FMC-DETR-B is better suited

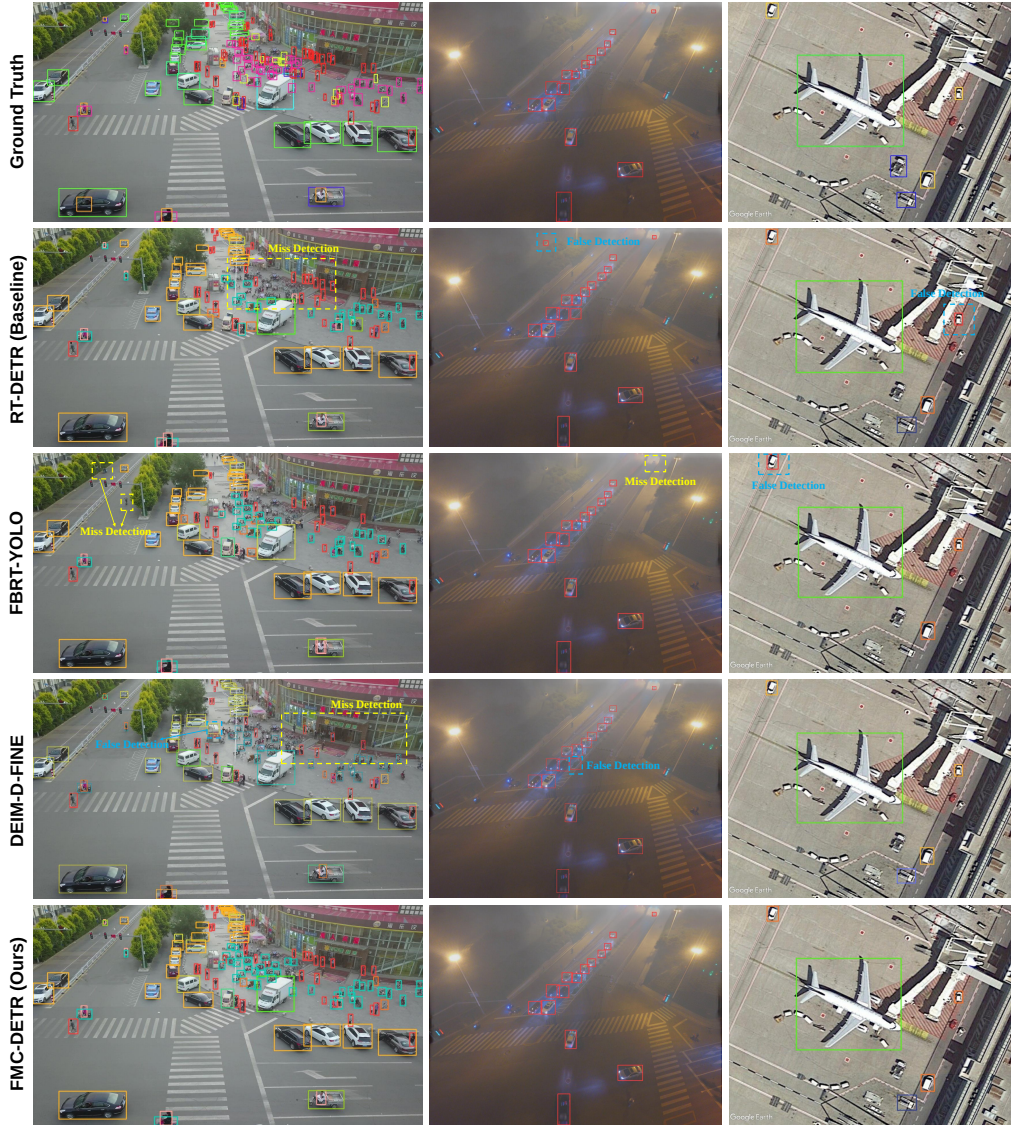


Fig. 7. Qualitative comparison of detection results produced by different methods across three aerial-view datasets. Each column corresponds to a different dataset, and the last row shows the results of the proposed FMC-DETR, which exhibits more accurate localization and fewer missed or false detections.

to accuracy-oriented aerial detection scenarios, where detection performance is prioritized over strict real-time latency. The additional latency indicates that hardware-aware acceleration and operator-level optimization remain important future directions, especially for the KAN-related components that are not yet as efficiently supported as standard convolutional operations in existing inference frameworks. Therefore, we plan to improve the practical efficiency of FMC-DETR-B through lightweight architectural redesign, operator re-parameterization, and deployment-oriented optimization. In future work, we plan to improve the practical efficiency of FMC-DETR-B through lightweight architectural redesign, operator re-parameterization, and deployment-oriented optimization.

V. CONCLUSION

In this paper, we propose FMC-DETR, a remote sensing object detector built upon frequency-aware modeling and multi-domain feature coordination. Specifically, we first design the

WeKat backbone to enhance shallow structural representations and strengthen deep nonlinear feature modeling. Then, we introduce the MDFC module to coordinate spatial, frequency, and structural cues, enabling the network to refine multi-domain feature correlations and extract more discriminative small-object representations. Finally, we develop the CPF module to progressively refine informative spatial responses and aggregate diverse multi-branch features, enhancing local contextual modeling while maintaining stable feature propagation. Extensive qualitative and quantitative experiments validate the effectiveness of FMC-DETR, demonstrating that it achieves competitive performance, without relying on any pretrained weights or additional training strategies.

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